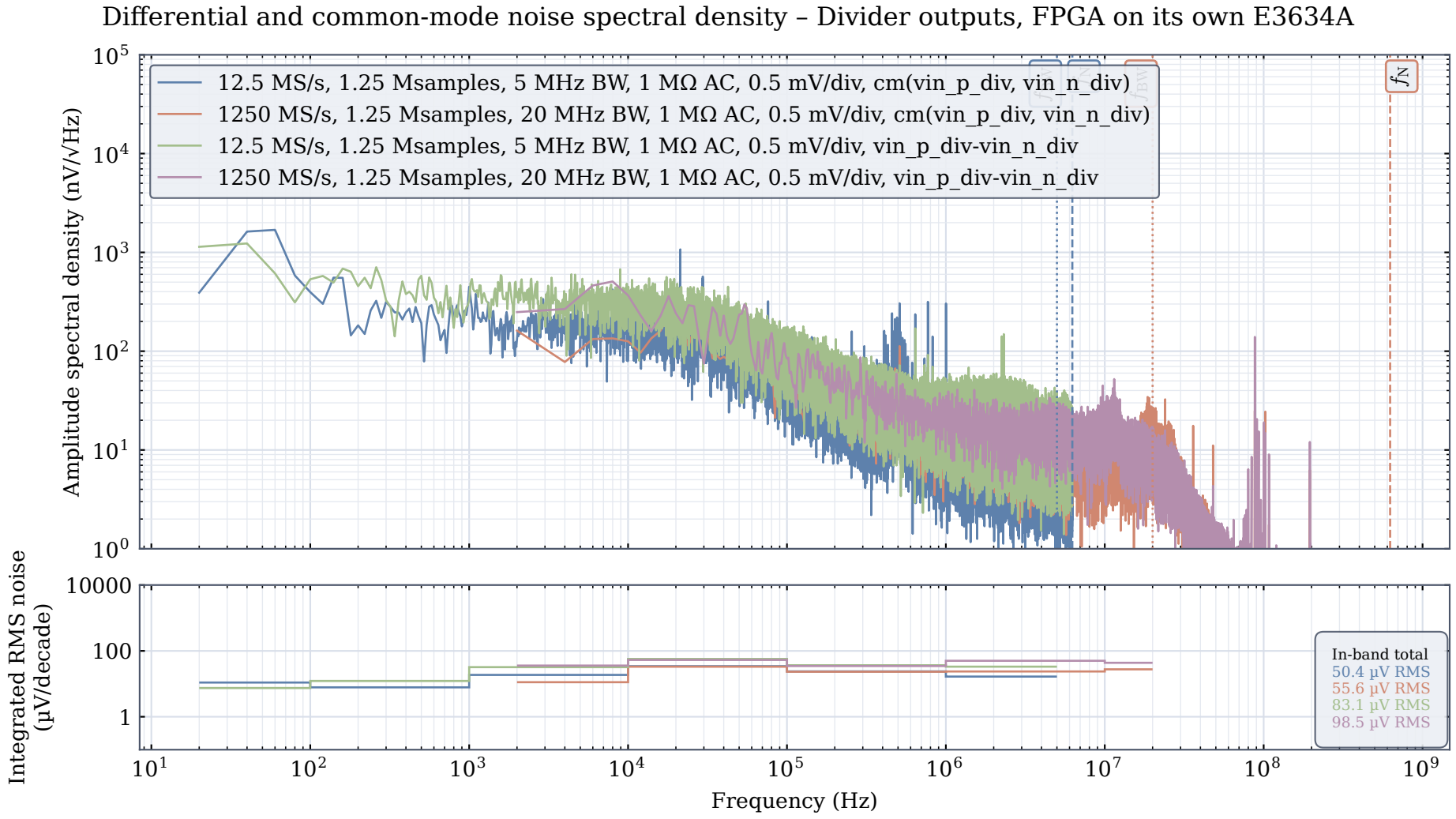

Differential and common-mode noise spectral density – Divider outputs, FPGA on its own E3634A
12.5 MS/s, 1.25 Msamples, 5 MHz BW, 1 MΩ AC, 0.5 mV/div, cm(vin_p_div, vin_n_div)
1250 MS/s, 1.25 Msamples, 20 MHz BW, 1 MΩ AC, 0.5 mV/div, cm(vin_p_div, vin_n_div)
12.5 MS/s, 1.25 Msamples, 5 MHz BW, 1 MΩ AC, 0.5 mV/div, vin_p_div-vin_n_div
1250 MS/s, 1.25 Msamples, 20 MHz BW, 1 MΩ AC, 0.5 mV/div, vin_p_div-vin_n_div
f_N
Amplitude spectral density (nV/√Hz)
10^5
10^4
10^3
10^2
10^1
10^0
Integrated RMS noise (µV/decade)
10000
100
1
In-band total
50.4 µV RMS
55.6 µV RMS
83.1 µV RMS
98.5 µV RMS
10^1
10^2
10^3
10^4
10^5
10^6
10^7
10^8
10^9
Frequency (Hz)